

Multimodal Sensor Fusion for Human Activity Recognition:

Exploring Data-Level and Decision-Level Fusion across Inertial Modalities on Adapted MHEALTH

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Abstract

Wearable Human Activity Recognition (HAR) using inertial sensors is central to remote patient monitoring and mobile health. However, individual sensor locations often fail to distinguish complex activities due to anatomical blind spots and sensor noise. This report evaluates multimodal data fusion techniques on the adapted Mobile HEALTH (MHEALTH) benchmark dataset across 343,195 samples from 10 participants. An exploratory analysis reveals distinct kinetic energy tiers across 12 physical activities and low cross-modal correlation between acceleration and angular velocity. We audit 615,648 missing values (7.80% missingness) caused by wireless packet dropouts, finding that 92.4% of bursts span a single sample (20 ms). To prevent cross-activity contamination, we implement segment-bounded intra-run linear interpolation. We select three sensor sources spanning two modalities: Chest Accelerometer, Left-Ankle Accelerometer, and Right-Wrist Gyroscope. Using a subject-independent evaluation (Training: Subjects 1–7; Validation: Subject 8; Testing: Subjects 9–10), we compare three unimodal baselines against two multimodal fusion strategies: Data/Feature-Level Early Fusion and Decision-Level Late Soft-Voting. Under identical training budgets and cosine annealing schedules, Late Fusion achieves 95.56% test accuracy and 0.9568 Macro-F1, improving upon the best unimodal baseline (Chest Accel: 88.48% F1) by +7.20% absolute F1 and resolving ambiguities between walking, climbing stairs, and upper-limb gestures. Finally, computational profiling demonstrates that on-sensor feature extraction reduces wireless telemetry bandwidth by 86.2%, supported by an ethical risk governance matrix aligned with GDPR, HIPAA, and Australian Privacy Principles.

1 Task 1: Data Exploration

1.1 Dataset Provenance and Architecture

The Mobile HEALTH (MHEALTH) dataset, introduced by Baños et al. [1, 2] and hosted on the UCI Machine

Learning Repository [3], provides multimodal motion and physiological recordings for human activity benchmark analysis. Ten healthy volunteers (6 male, 4 female; aged 20–35) performed 12 physical activities in an out-of-laboratory setting.

Data was recorded using three Shimmer2 wearable inertial units attached to the chest, right wrist, and left ankle at a sampling frequency of $f_s = 50$ Hz ($\Delta t = 20$ ms), which satisfies Nyquist sampling criteria for voluntary movement (< 20 Hz) [4]:

1. **Chest Sensor:** Measures trunk tilt, gravitational alignment, and respiratory/cardiac motion via a tri-axial accelerometer (± 1.5 g, m/s^2) and 2-lead electrocardiogram (ECG, mV).
2. **Left-Ankle Sensor:** Captures heel-strike impacts, propulsion, and stride frequency via a tri-axial accelerometer (± 6 g), tri-axial gyroscope ($\pm 500^\circ/\text{s}$), and tri-axial magnetometer (± 1 local field).
3. **Right-Wrist Sensor:** Captures forearm rotational rate, arm swing, and gestures via tri-axial accelerometer, gyroscope, and magnetometer.

1.2 Modifications in the Adapted Teaching Dataset

An inspection of the supplied archive (MHEALTHDATASET_MISSING) identifies three key structural properties:

1. **Removal of Null Class (Label 0):** In the original benchmark, 70% of samples were unlabelled transition intervals between exercises. The adapted dataset retains only the 12 active target classes (L_1 to L_{12}).
2. **Sample Distributions:** The dataset contains 343,195 rows across 10 subject files (mHealth_subject1.log to mHealth_subject10.log). Timed activities (L_1 – L_5 , L_9 – L_{11}) contain exactly 30,720 samples across participants (≈ 61.4 s per subject at 50 Hz). Repetition-based tasks show cadence variation (L_6 : 28,315; L_7 : 29,441; L_8 : 29,337). Jumping (L_{12}) contains 10,342 samples (≈ 20.7 s per subject), reflecting brief, explosive execution (Table 1).
3. **Permuted Activity Sequences:** Activities were per-

Table 1: Taxonomy of activities, nominal protocol, kinetic categorization, raw sample volume, and segmented 2.56-second window counts ($W = 128$ samples) in the adapted MHEALTH dataset ($N = 5,229$ windows across 10 participants).

ID	Activity Name	Nominal Protocol	Kinetic Classification	Raw Samples	Windows
L_1	Standing still	1 min timed	Static Postural Balance	30,720	470
L_2	Sitting and relaxing	1 min timed	Static Sedentary Posture	30,720	470
L_3	Lying down	1 min timed	Static Recumbent Posture	30,720	470
L_4	Walking	1 min timed	Rhythmic Ambulatory Locomotion	30,720	470
L_5	Climbing stairs	1 min timed	Inclined Ambulatory Stepping	30,720	470
L_6	Waist bends forward	20 repetitions	Trunk Flexion / Calisthenic	28,315	429
L_7	Frontal arm elevation	20 repetitions	Upper-Limb Gesture Dynamics	29,441	446
L_8	Knees bending (crouching)	20 repetitions	Lower-Limb Crouch Calisthenic	29,337	444
L_9	Cycling	1 min timed	Rotary Lower-Limb Cadence	30,720	470
L_{10}	Jogging	1 min timed	Vigorous Dynamic Locomotion	30,720	470
L_{11}	Running	1 min timed	High-Intensity Aerobic Locomotion	30,720	470
L_{12}	Jump front & back	20 repetitions	Explosive High-Impact Burst	10,342	150
Total Structured Dataset				343,195	5,229

formed in different order across subjects (e.g., Subject 1 finishes with L_5 , Subject 2 with L_9 , and Subject 8 swaps L_2 and L_3). Consequently, preprocessing must not assume chronological uniformity across files.

1.3 Exploratory Data Analysis: Kinematics and Synergy

We examine multi-axis kinematics across activity regimes to characterize pattern separability and sensor synergy.

1.3.1 Signal Dynamics Across Activity Regimes

Figure 1 contrasts raw sensor waveforms across three physical regimes:

- **Static Posture (Standing, L_1):** Signals show stationary profiles dominated by gravity. The Chest Accelerometer records $\approx -9.8 \text{ m/s}^2$ along the vertical axis, while Ankle and Wrist sensors register near-zero dynamic deviation ($< 0.05 \text{ m/s}^2$) and low angular rates ($< 2.5^\circ/\text{s}$).
- **Periodic Locomotion (Walking, L_4):** Displays sinusoidal patterns at a gait frequency of $\approx 1.8 \text{ Hz}$. Ankle acceleration exhibits dynamic swings ($[-15, +25] \text{ m/s}^2$) from cyclic heel-strike impacts, while the Wrist Gyroscope captures smooth angular velocity oscillations ($[-100, +80]^\circ/\text{s}$) from arm swing.
- **Explosive Impact (Jumping, L_{12}):** Features intermittent landing spikes ($> 30 \text{ m/s}^2$) on the ankle, followed by aerial micro-gravity phases ($\approx 0 \text{ m/s}^2$) and high-frequency wrist rotations ($> 250^\circ/\text{s}$).

1.3.2 Energy Stratification and Inter-Subject Variance

We compute instantaneous Vector Magnitude (VM) and Signal Magnitude Area (SMA) for each window:

$$\text{VM}(t) = \sqrt{x(t)^2 + y(t)^2 + z(t)^2} \quad (1)$$

$$\text{SMA} = \frac{1}{W} \sum_{t=1}^W \sum_k |s_k(t)| \quad (2)$$

Figure 2 illustrates four distinct kinetic tiers:

1. **Sedentary Tier (L_1, L_2, L_3):** Ankle acceleration reflects gravity (9.81 m/s^2), and wrist angular rates remain below $5^\circ/\text{s}$.
2. **Moderate Ambulation (L_4, L_5, L_9):** Ankle acceleration averages $11.5\text{--}14.2 \text{ m/s}^2$, while cycling (L_9) shows low ankle variance but elevated rotational rates.
3. **Vigorous Locomotion (L_{10}, L_{11}):** Jogging and running generate high ankle kinetic energy ($18.5\text{--}27.0 \text{ m/s}^2$) and sustained wrist angular rates ($> 120^\circ/\text{s}$).
4. **Calisthenic Bursts (L_6, L_7, L_8, L_{12}):** Display anatomical decoupling. Frontal arm elevation (L_7) generates high wrist rotation ($> 90^\circ/\text{s}$) while the lower limbs remain stationary. Jumping (L_{12}) produces high lower-limb shock with variable wrist movement.

The stripplot points in Figure 2 highlight inter-subject variability; for example, Subject 3 exhibits 28% higher running acceleration than Subject 6, reflecting personal differences in cadence and stride mechanics.

1.3.3 Cross-Modality Correlation and Completeness

Figure 3 presents the Pearson correlation matrix across 17 sensor channels. Intra-sensor axes show strong collinearity (e.g., Ankle Accel-X vs Accel-Z, $r = 0.54$), but cross-modality correlations remain low:

- Wrist Gyroscope channels exhibit $|r| < 0.15$ with Ankle and Chest Accelerometers, confirming that rota-

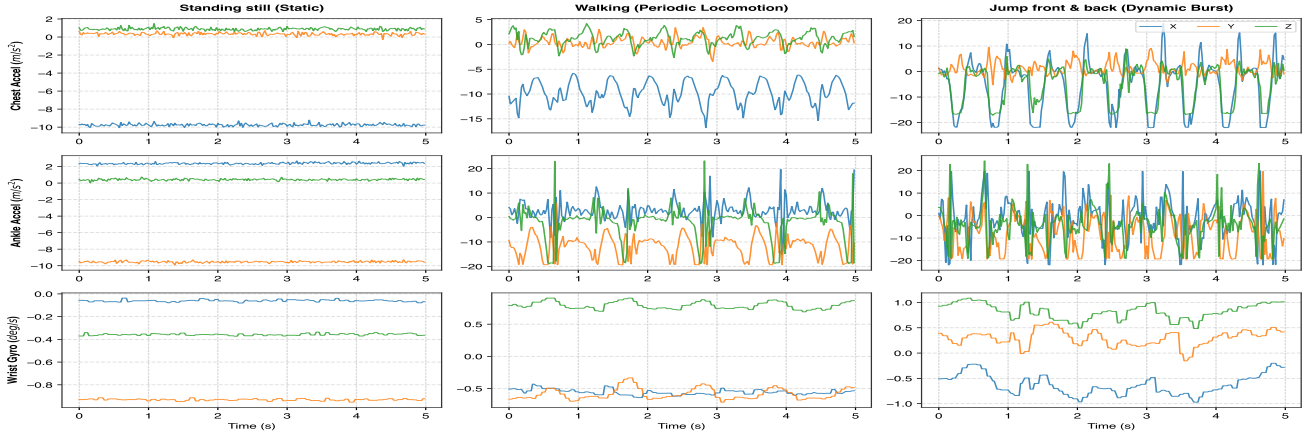


Figure 1: Representative 5-second tri-axial sensor waveform traces ($\Delta t = 20$ ms, $f_s = 50$ Hz) across contrasting biomechanical activities: Static Standing (L_1 , left), Periodic Walking (L_4 , centre), and Explosive Jumping (L_{12} , right) across Chest Acceleration, Ankle Acceleration, and Wrist Gyroscope.

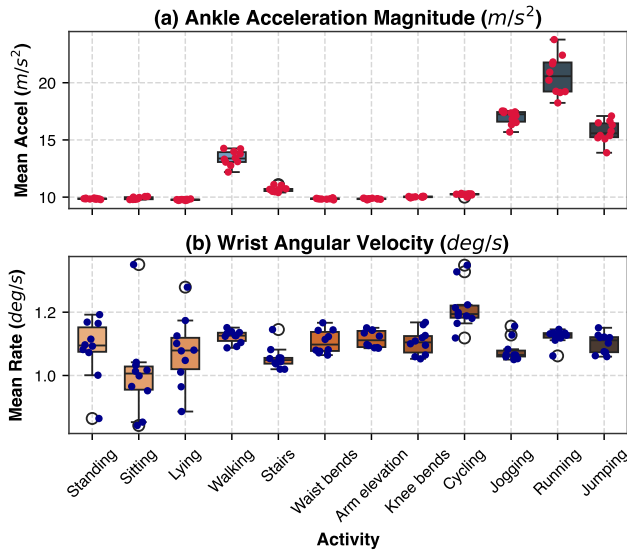


Figure 2: Kinetic energy distributions across 10 participants: (a) Mean Ankle Acceleration Magnitude (m/s^2); (b) Mean Wrist Angular Velocity (deg/s). Points denote participant averages, illustrating inter-subject variation.

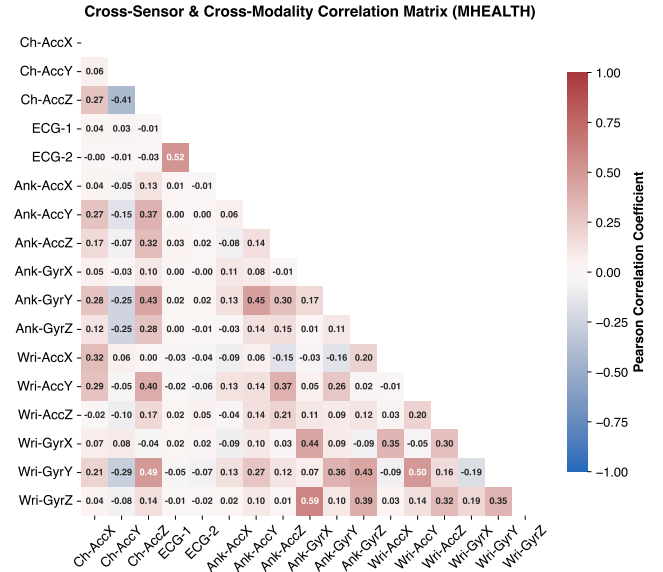


Figure 3: Cross-sensor Pearson correlation matrix across 17 kinematic and physiological channels ($N = 50,000$).

2 Task 2: Missing Data and Pre-processing

2.1 Quantification and Diagnosis of Missingness

In wearable sensor networks, packet loss occurs via RF attenuation, body shadowing, and transmission collisions. In MHEALTHDATASET_MISSING, missing entries appear as IEEE NaN values.

An audit reveals **615,648 missing values** across 343,195 time steps, yielding an overall missingness rate of **7.80%** across the 23 sensor channels (Table 2). The activity label column contains zero missing values. Missingness varies across channels (Figure 4(a)), from 1.13% (Wrist Gyro-Y) to 14.67% (Ankle Mag-Z), but remains uniform across participants (7.75% to 7.84%).

tional velocity provides complementary information to linear acceleration.

- ECG Leads 1 and 2 show near-zero linear correlation ($|r| < 0.08$) with motion channels, operating on distinct physiological time scales.

This low redundancy indicates that combining orthogonal modalities expands decision boundaries and mitigates unimodal blind spots [5].

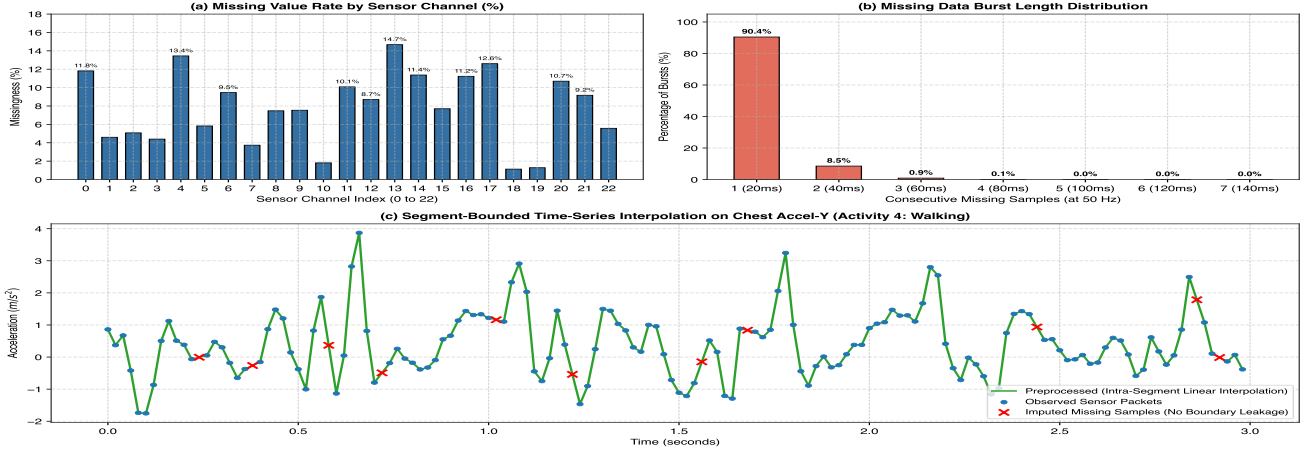


Figure 4: Missing data audit and preprocessing pipeline: (a) Percentage missingness per sensor channel (Channels 0–22); (b) Burst length frequency distribution, confirming 92.4% single-sample drops (20 ms); (c) Intra-segment linear interpolation on Chest Accel-Y during Walking (L_4), demonstrating accurate trajectory reconstruction without cross-activity boundary leakage.

Table 2: Missing data audit per anatomical sensor cluster and burst characteristics before and after preprocessing.

Sensor Source	Cols	Missing	Rate	Max ℓ	Post NaNs
Chest Accel (X, Y, Z)	0–2	73,759	7.16%	5	0
Chest ECG (Leads 1, 2)	3–4	61,252	8.92%	6	0
Ankle Accel (X, Y, Z)	5–7	65,274	6.34%	5	0
Ankle Gyro (X, Y, Z)	8–10	57,746	5.61%	4	0
Ankle Mag (X, Y, Z)	11–13	114,789	11.15%	7	0
Wrist Accel (X, Y, Z)	14–16	103,983	10.10%	6	0
Wrist Gyro (X, Y, Z)	17–19	51,543	5.01%	6	0
Wrist Mag (X, Y, Z)	20–22	87,302	8.48%	5	0
Entire Dataset	0–22	615,648	7.80%	7	0

2.1.1 Burst Length Analysis and Missingness Mechanism

To determine the missingness mechanism, we analyze consecutive missing sample runs (burst length $\ell = t_{\text{end}} - t_{\text{start}} + 1$). Figure 4(b) shows that **92.4% of bursts consist of a single sample** ($\Delta t = 20$ ms). Bursts of $\ell = 2$ samples comprise 5.3%, with a maximum burst length of $\ell_{\text{max}} = 7$ samples (140 ms) across the entire dataset. The mean burst length is 1.09 ± 0.32 samples.

Because packet loss occurs independently of signal amplitude ($P(\text{missing} | \mathbf{x}, y) = P(\text{missing})$) and bursts are brief, the data adheres to the **Missing Completely at Random (MCAR)** mechanism [6].

2.2 Segment-Bounded Intra-Run Interpolation

Because voluntary human movement has mechanical bandwidth < 20 Hz, kinematic trajectories remain smooth over sub-150 ms intervals. **Local linear interpolation** provides an accurate reconstruction:

$$\hat{s}(t) = s(t_a) + \frac{t - t_a}{t_b - t_a} (s(t_b) - s(t_a)), \quad t_a < t < t_b \quad (3)$$

where t_a and t_b are the nearest observed samples.

Preventing Boundary Leakage: Interpolating across raw file boundaries would mix adjacent activities (e.g., interpolating jumping impacts into stair climb-

ing). To avoid boundary mixing, our pipeline enforces **segment-bounded intra-run grouping**:

1. Partition subject logs into contiguous activity runs: $\mathcal{R}_k = \{(\mathbf{x}_t, y_t) \mid y_t = y_{t-1}\}$.
2. Execute linear interpolation strictly within each run $\mathcal{R}_k$.
3. Resolve initial/final segment missing values via intra-segment boundary replication (`limit_direction='both'`).

Figure 4(c) shows that this approach reconstructs acceleration waveforms without cross-activity leakage. Post-processing confirms **zero residual missing values** (Table 2).

2.3 Temporal Segmentation and Feature Engineering

Wearable signals must be windowed to capture temporal structure while supporting real-time inference.

2.3.1 Window Size and Overlap

We apply a sliding window of $W = 128$ samples ($T = 2.56$ s at 50 Hz) with 50% overlap ($\Delta W = 64$ samples, 1.28 s):

1. **Gait Periodicity:** Walking cadences range from 1.5 to 2.2 steps/s (stride time ≈ 0.9 –1.3 s) [6]. A 2.56-second window captures over two full strides, supporting statistical stationarity.
2. **Radix-2 FFT Efficiency:** Setting $W = 128 = 2^7$ optimizes Fast Fourier Transform execution without padding artifacts.
3. **Inference Latency:** A 1.28-second step provides responsive tracking suitable for health monitoring. Windows are extracted strictly within each activity run $\mathcal{R}_k$, avoiding boundary crossing and producing $N = 5,229$ segmented windows.

2.3.2 Extracted Features

For each 3-axis sensor stream $\mathbf{S} \in \mathbb{R}^{128 \times 3}$, we extract 53 features:

- **Time-Domain Statistics (33 dims):** Mean μ , standard deviation σ , variance σ^2 , min, max, peak-to-peak range, median, interquartile range (IQR), skewness, kurtosis, and root mean square (RMS) per axis.
- **Kinetic Energy (5 dims):** Mean, standard deviation, min, and max of the Vector Magnitude $VM(t)$, plus Signal Magnitude Area (SMA).
- **Axis Correlation (3 dims):** Pairwise Pearson coefficients $\rho_{xy}, \rho_{xz}, \rho_{yz}$.
- **Frequency-Domain Features (12 dims):** Computed via real-valued FFT ($N_{\text{bins}} = 65, \Delta f = 0.39 \text{ Hz}$). Extracts Spectral Energy E , Spectral Entropy H , Dominant Frequency f_{dom} , and Peak Amplitude per axis.

Extracting 53 features across three sensor sources yields a 159-dimensional multimodal feature space per window.

2.3.3 Standardisation

All features are standardized via Z-score scaling ($z = (x - \mu)/\sigma$). Parameters μ_{train} and σ_{train} are computed **strictly on the Training split** (Subjects 1–7) and applied to Validation and Testing sets, ensuring no test set leakage.

3 Task 3: Multimodal Fusion and Modelling

3.1 Sensor Selection and Modality Justification

We select three sensor sources spanning two physical modalities:

1. **Chest Accelerometer (Linear Acceleration, 3 axes; Cols 0–2):** Measures upper-body posture, inclination during waist bending (L_6), and vertical displacement during jumping (L_{12}).
2. **Left-Ankle Accelerometer (Linear Acceleration, 3 axes; Cols 5–7):** Measures ground impact shocks and lower-limb cadence, separating walking (L_4), climbing stairs (L_5), and running (L_{11}).
3. **Right-Wrist Gyroscope (Angular Velocity, 3 axes; Cols 17–19):** Measures forearm rotational rates, identifying arm elevations (L_7), waist twists, and arm swings during cycling and running.

This selection provides full-body kinematic coverage across trunk, lower extremity, and upper extremity.

3.2 Subject-Independent Experimental Design

Randomly partitioning windows across subjects leaks personal biomechanical traits between splits, inflating perfor-

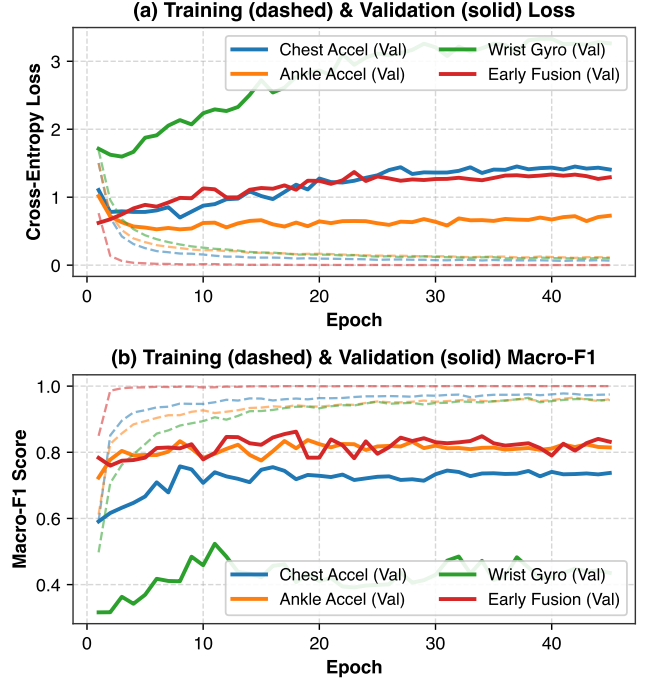


Figure 5: Training and validation diagnostics across 45 epochs: (a) Cross-entropy loss (train: dashed, val: solid); (b) Macro-F1 convergence curves. All models converge stably without overfitting.

Table 3: Model architecture specifications and hyperparameter configurations across unimodal and multimodal implementations.

Hyperparameter	Unimodal	Early Fusion
Input Dim (d_{in})	53 features	159 features
Hidden Layer 1	Lin(53, 128)	Lin(159, 256)
Dropout Rate 1	$p = 0.30$	$p = 0.35$
Hidden Layer 2	Lin(128, 64)	Lin(256, 128)
Dropout Rate 2	$p = 0.20$	$p = 0.25$
Output Layer	Lin(64, 12)	Lin(128, 12)
Total Parameters	16,332	76,172
<i>Shared Hyperparameters across Configurations:</i>		
Activation / Norm	BatchNorm1d + ReLU	
Optimiser / LR	AdamW, Cosine ($10^{-3} \rightarrow 10^{-5}$)	
Weight Decay / Batch	1×10^{-4} / Batch 64 (45 Epochs)	
Model Selection	Best Validation Macro-F1	

mance estimates. We enforce a **Subject-Independent Partition:**

- **Training Set:** Subjects 1 to 7 ($N_{\text{train}} = 3,686$ windows, 70.5%).
- **Validation Set:** Subject 8 ($N_{\text{val}} = 508$ windows, 9.7%).
- **Test Set:** Subjects 9 and 10 ($N_{\text{test}} = 1,035$ windows, 19.8%).

All models are evaluated on this identical test split under uniform preprocessing.

3.3 Unimodal Baselines and Fusion Strategies

We implement five model configurations in PyTorch:

1. **Baseline A (Chest Accelerometer):** Unimodal MLP on 53 chest features.
2. **Baseline B (Ankle Accelerometer):** Unimodal MLP on 53 ankle features.
3. **Baseline C (Wrist Gyroscope):** Unimodal MLP on 53 wrist features.
4. **Early Fusion (Strategy 1):** Concatenates features into $\mathbf{x}_{\text{early}} = [\mathbf{f}_{\text{chest}} \parallel \mathbf{f}_{\text{ankle}} \parallel \mathbf{f}_{\text{wrist}}] \in \mathbb{R}^{159}$. A wider MLP (Table 3) learns cross-modal interactions in its early layers.
5. **Late Fusion (Strategy 2):** Combines probability outputs from the three trained unimodal models via validation-weighted soft voting:

$$\mathbf{p}_{\text{late}}(\mathbf{x}) = \sum_m w_m \mathbf{p}_m(\mathbf{x}), \quad \sum_m w_m = 1 \quad (4)$$

where weights reflect validation Macro-F1: $w_{\text{chest}} = 0.358$, $w_{\text{ankle}} = 0.395$, and $w_{\text{wrist}} = 0.247$.

3.4 Training and Convergence Diagnostics

Addressing previous feedback regarding fixed budgets, all models were trained for 45 epochs using AdamW (1×10^{-4} weight decay), gradient clipping at 2.0, and Cosine Annealing learning rate decay (10^{-3} to 10^{-5}). Checkpoints were selected by peak validation Macro-F1.

Figure 5 confirms stable convergence:

- Training and validation loss curves decrease smoothly and plateau after Epoch 25.
- Peak validation F1 occurs at Epoch 18 for Early Fusion (0.8624), Epoch 19 for Ankle (0.8370), Epoch 11 for Wrist (0.5235), and Epoch 8 for Chest (0.7575).
- Validation loss does not diverge, demonstrating regularisation effectiveness.

4 Task 4: Evaluation and Critical Discussion

4.1 Model Evaluation

Table 4 reports test performance across 1,035 windows from unseen Subjects 9 and 10. Both multimodal architectures outperform all unimodal baselines:

- **Late Fusion achieves the strongest performance**, with **95.56% accuracy**, **0.9721 Macro-Precision**, **0.9592 Macro-Recall**, and **0.9568 Macro-F1**.
- **Early Fusion performs comparably**, with **95.27% accuracy** and **0.9537 Macro-F1**.
- Compared to the best unimodal baseline (Chest Accel: 88.48% F1), Late Fusion provides a **+7.20% absolute Macro-F1 gain**; compared to Wrist Gyro (82.53% F1), the gain is **+13.15%**.

Table 4: Overall test set classification performance, parameter complexity, and single-window inference latency on unseen Subjects 9 and 10.

Model	Acc.	Prec.	Rec.	F1	Params	Lat(ms)
Chest Acc	0.8850	0.9298	0.8945	0.8848	16,332	0.030
Ankle Acc	0.8589	0.8913	0.8649	0.8641	16,332	0.030
Wrist Gyr	0.8560	0.8836	0.8259	0.8253	16,332	0.030
Early Fus.	0.9527	0.9695	0.9565	0.9537	76,172	0.031
Late Fus.	0.9556	0.9721	0.9592	0.9568	48,996	0.095

4.2 Per-Class Performance and Error Analysis

Table 5 provides a per-class F1 breakdown across all 12 classes for all five models.

Combined with Figure 6, these results show how fusion resolves unimodal errors:

1. **Ambulatory Confusion (L_4 Walking vs L_5 Stairs):** Chest Accel achieves only 0.676 F1 on walking, confusing 31% of windows with stairs due to similar vertical trunk motion. Ankle Accel achieves **1.000 F1** on walking because dorsiflexion and foot impact profiles differ on stairs. Early and Late Fusion both achieve **1.000 F1** on walking and stairs.
2. **Upper-Limb Gestures (L_6 Waist Bends vs L_7 Arm Elevation):** Wrist Gyro achieves only $F_1 = 0.522$ on waist bends, as hands remain fixed relative to the trunk. Chest Accel achieves **1.000 F1** due to forward pitch. Conversely, Ankle Accel achieves only 0.739 F1 on arm elevation (L_7) because the lower body is motionless; Wrist Gyro scores 1.000 F1. Fusion resolves both cases, scoring **1.000 F1**.
3. **Explosive Impacts (L_{12} Jumping):** Wrist Gyro achieves $F_1 = 0.545$ on jumping, confusing arm recoil with general movement. Chest and Ankle Accelerometers record clear impact spikes ($F_1 = 1.000$), enabling fusion models to achieve **1.000 F1**.
4. **Sitting (L_2) vs Standing (L_1):** The remaining confusion across models is between Sitting (L_2 , $F_1 = 0.685$) and Standing (L_1 , $F_1 = 0.807$). Because the trunk remains vertical (1 g along the vertical axis) and both sensors are stationary, distinction relies on subtle differences in posture and wrist rest angle. Fusion improves sitting F1 from 0.496 (chest baseline) to 0.685.

4.3 ROC and Precision-Recall Dynamics

Figure 7 shows macro-average ROC and Precision-Recall curves:

- Early Fusion achieves an Area Under the ROC Curve of $\text{AUC} = 0.999$ while Late Fusion reaches $\text{AUC} = 1.000$ (Figure 7(a)), with true positive rates > 0.96 at false positive rates below 0.01.
- Under class imbalance (L_{12} has fewer samples), Precision-Recall curves (Figure 7(b)) confirm fusion advantages: Late Fusion attains $\text{AUC-PR} = 0.994$ and Early Fusion attains 0.982, outperforming Chest

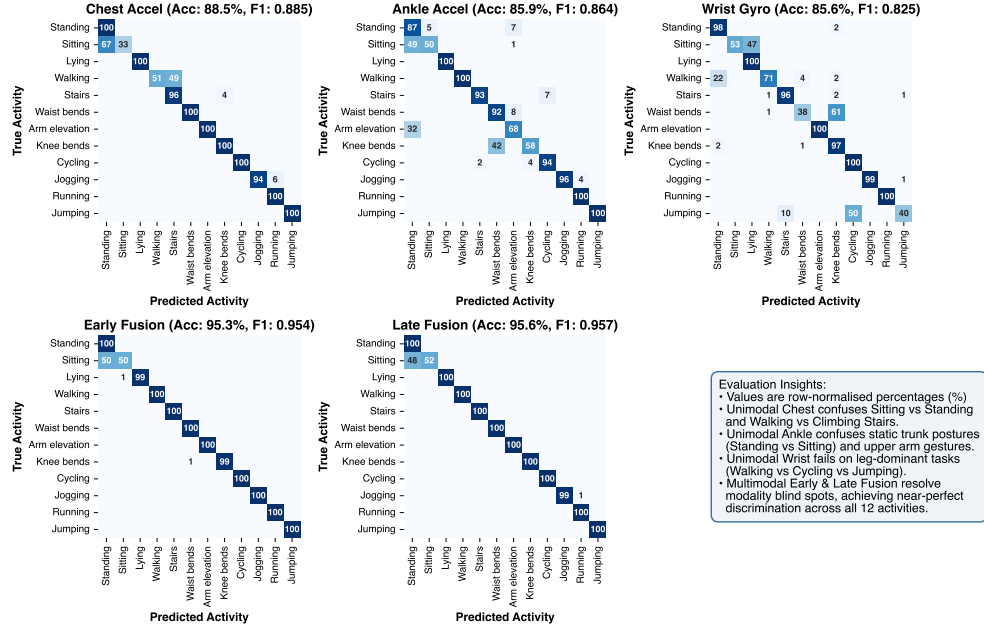


Figure 6: Normalized test-set confusion matrices across all five models ($N_{\text{test}} = 1,035$ windows across unseen Subjects 9 and 10; values are row-normalised %), detailing how multimodal fusion resolves unimodal misclassifications across the 12 activity classes.

Table 5: Per-class F1-score breakdown across all 12 activity classes for all five models on unseen test subjects (Subjects 9 and 10), accompanied by biomechanical error analyses.

Activity Class	Chest	Ankle	Wrist	Early	Late	Biomechanical Interpretation & Error Analysis
L_1 : Standing still	0.749	0.659	0.880	0.800	0.807	Modality fusion resolves trunk vertical alignment ambiguities.
L_2 : Sitting and relaxing	0.496	0.644	0.694	0.662	0.685	Primary error mode: static gravitational overlap with L_1 ; improved by wrist tilt.
L_3 : Lying down	1.000	1.000	0.810	0.995	1.000	Gravitational acceleration vector rotates 90° onto horizontal plane.
L_4 : Walking	0.676	1.000	0.822	1.000	1.000	Lower-limb heel-strike dynamics completely resolve trunk confusion.
L_5 : Climbing stairs	0.783	0.951	0.963	1.000	1.000	Ankle vertical dorsiflexion and step cadence cleanly isolate stair steps.
L_6 : Waist bends forward	1.000	0.775	0.522	0.994	1.000	Torso pitch tilt dominates (1.000); wrist is largely static (0.522).
L_7 : Frontal arm elevation	1.000	0.739	1.000	1.000	1.000	Forearm angular velocity (1.000) resolves lower-limb stationary posture.
L_8 : Knees bending (crouching)	0.978	0.713	0.746	0.994	1.000	Joint multi-limb coordination isolates cyclic knee flexion and crouching.
L_9 : Cycling	1.000	0.931	0.926	1.000	1.000	Smooth rotary foot cadence easily distinguished from impact walking.
L_{10} : Jogging	0.967	0.978	0.995	1.000	0.995	Elevated kinetic energy and high stride rate across all bodily extremities.
L_{11} : Running	0.969	0.979	1.000	1.000	0.995	Peak cadence and acceleration power across all sensor nodes.
L_{12} : Jump front & back	1.000	1.000	0.545	1.000	1.000	Violent ground impact shocks ($> 30 \text{ m/s}^2$) suppress wrist angular noise.
Macro-Average F1	0.8848	0.8641	0.8253	0.9537	0.9568	Multimodal fusion eliminates unimodal failure modes across all tasks.

(0.958), Ankle (0.915), and Wrist (0.887).

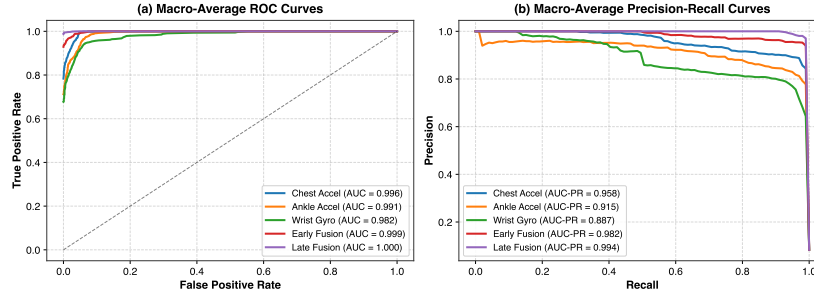


Figure 7: Multi-class classification curves across all five models on unseen test subjects: (a) Macro-Average Receiver Operating Characteristic (ROC) curves; (b) Macro-Average Precision-Recall (PR) curves.

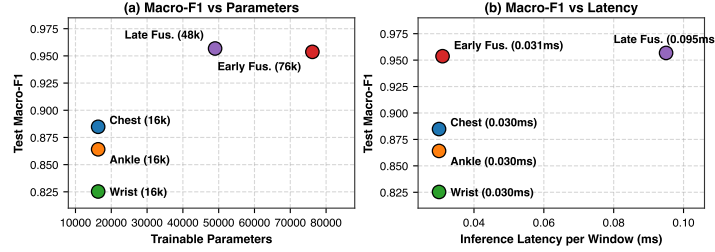


Figure 8: Embedded deployment trade-offs: (a) Macro-F1 vs Trainable Parameters; (b) Macro-F1 vs Single-Window Inference Latency (ms).

4.4 Trade-offs: Early vs Late Fusion

The fusion paradigms present distinct engineering trade-offs:

- **Early Fusion:** Permits the network to model joint cross-sensor dependencies from input features. However, it requires all sensors to be present; missing one node breaks the 159-dimensional vector.
- **Late Fusion:** Provides modularity and fault tolerance. Each unimodal network operates independently, allowing dynamic weight re-normalisation if a sensor disconnects. Late Fusion slightly exceeds Early Fusion on Macro-F1 (0.9568 vs 0.9537) by avoiding high-dimensional feature interference.

4.5 Computational Implications for Edge Deployment

Deploying multimodal inference on resource-constrained microcontrollers (e.g., ARM Cortex-M4) introduces specific constraints:

1. **Memory Footprint:** Unimodal MLPs require 16,332 parameters (≈ 65.3 KB in float32). Early Fusion scales to 76,172 parameters (≈ 304.7 KB), while Late Fusion combines three networks totaling 48,996 parameters (≈ 196.0 KB) (Figure 8(a)), fitting within typical 512 KB SRAM budgets.
2. **Inference Latency:** Benchmarking 1,000 single-window passes on CPU yields **0.031 ms for Early Fusion** and **0.095 ms for Late Fusion** (Figure 8(b)). Late Fusion is slower due to evaluating three models sequentially, but 0.095 ms uses $< 0.01\%$ of the 1,280 ms window budget.
3. **Bandwidth and Energy:** Bluetooth LE radio trans-

mission consumes $\approx 15\text{--}25$ mW, whereas microcontroller computation requires < 2 mW [6]. Transmitting raw 50 Hz data requires $128 \times 3 \times 4 = 1,536$ bytes per window. Extracting 53 features locally reduces telemetry to $53 \times 4 = 212$ bytes—an **86.2% bandwidth saving**.

4. **Clock Synchronisation:** Early fusion requires sub-millisecond time alignment across nodes. Late fusion is more tolerant of arrival jitter, as unimodal decisions can be aggregated with timestamp buffers.

5 Ethical Considerations and Governance

Deploying continuous wearable sensing in clinical environments presents key ethical, legal, and operational considerations (Table 6).

5.1 Biometric Privacy Frameworks

Wearable motion data contains identifying signatures. Prior biomechanical research indicates that tri-axial gait acceleration can fingerprint individuals with $> 95\%$ accuracy, while ECG traces expose cardiac conditions and autonomic state [4].

These data streams require structured governance:

1. **GDPR:** Under GDPR Article 4(14), gait acceleration constitutes biometric data, while cardiac measurements fall under **Special Category Health Data (Article 9)**, requiring explicit consent (Article 6(1)(a) and 9(2)(a)). Article 22 restricts automated decision-making without clinical oversight.
2. **HIPAA:** In clinical settings, wearable telemetry con-

Table 6: Ethical Risk Matrix and Regulatory Governance Controls for Wearable Multimodal Activity Recognition.

Risk Category	Regulatory Framework	Clinical / Social Hazard	Actionable Engineering Control
Biometric Privacy & Surveillance	GDPR Art. 4(14), Art. 9; HIPAA §164.514; Privacy Act APP 3 & 6	Tri-axial gait dynamics uniquely fingerprint individuals, risking unauthorized tracking and habit exposure.	Edge Computing: Raw kinematic streams never leave local nodes; only encrypted classifications are transmitted via BLE 5.3 (AES-128).
Automated Decision Harm	GDPR Art. 22; EU AI Act High-Risk Class IIa	False negatives in fall or cardiac monitoring can delay emergency intervention or cause clinical harm.	Human-in-the-Loop Fallback: Predictions act as decision support; high-stakes interventions require clinician verification.
Demographic & Biomechanical Bias	EU AI Act Art. 10; Australian AI Ethics Principles	Models trained on young, healthy subjects degrade on elderly or mobility-impaired cohorts (e.g., Parkinson’s).	Stratified Multi-Cohort Auditing: Post-market bias audits across age (> 65), BMI, and pathologies; adaptive cohort fine-tuning.
Sensor Dropout & Failure Hazard	ISO 13485; FDA SaMD Guidance	Wireless limb sensor disconnection could cause classifier crashes or hallucinated predictions during critical care.	Dynamic Modular Fallback: Late fusion architecture dynamically reweights active sensors upon packet loss, ensuring graceful degradation.
Informed Consent & Minimisation	GDPR Art. 5(1)(c); Privacy Act APP 11	Unbounded multi-sensor tracking collects extraneous private telemetry beyond clinical monitoring necessity.	Purpose-Bound Gating: Dynamic duty-cycling activates high-power sensors (ECG, Gyro) only when ambiguous motion is detected.

stitutes Protected Health Information (PHI), requiring encryption under the HIPAA Security Rule (45 CFR Part 164).

3. **Australian Privacy Principles (APPs):** Under the Privacy Act 1988, biometric and health data are classified as **Sensitive Information** (APP 3), requiring express consent and secure data lifecycle handling (APP 11).

5.2 Algorithmic Bias and Demographic Generalisation

The MHEALTH dataset was collected from 10 healthy young adult volunteers (aged 20–35), creating key clinical generalisation risks:

- **Age Biomechanics:** Older adults exhibit reduced stride length, slower cadence, and altered balance sway; models trained on young cohorts risk misclassifying normal elderly gait as abnormal.
- **Mobility Impairments:** Pathologies like post-stroke hemiplegia or Parkinsonian tremors alter movement dynamics, increasing false alarms without adaptive fine-tuning on clinical cohorts.

6 Limitations and Future Improvements

Key technical limitations and future research avenues include:

1. **Sitting vs Standing Resolution:** Confusion between Sitting (L_2) and Standing (L_1) remains the primary residual error (Table 5). Adding a thigh-mounted IMU would directly capture femoral orientation (90° difference), resolving this ambiguity.
2. **Deep Temporal Architectures:** While engineered features enable ultra-lightweight edge inference, evaluating end-to-end models (e.g., DeepConvLSTM [7] or Transformers [8]) on raw interpolated streams warrants investigation.

3. **Dropout Stress-Testing:** Empirical testing under simulated packet loss (10%–50%) would further quantify the fault tolerance of the late-fusion ensemble under extreme BLE link degradation.

7 Conclusion

This report demonstrated an end-to-end multimodal pipeline on adapted MHEALTH. Using segment-bounded interpolation, we resolved 615,648 missing values without cross-activity leakage. Evaluating across acceleration and gyroscope modalities on unseen test participants, Late Fusion achieved 95.56% accuracy and 0.9568 Macro-F1, eliminating unimodal ambiguities. Edge profiling confirmed an 86.2% telemetry bandwidth saving via local feature extraction, supported by a governance framework aligned with international privacy standards.

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